

Akbar Aman

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SUMMARY

Founder and Computer Engineering junior (May 2027) with 2+ years of experience in building full-stack & AI/ML systems across SaaS platforms, AI agents, and production web applications.

EDUCATION

University of Illinois Chicago, College of Engineering Expected: May 2027
Bachelor of Science – Computer Engineering, Minor – Mathematics GPA: 3.65

- **Leadership:** Computer Science TA (CS 109, 2 terms) & Engineering Success Mentor (ENGR 101, 2 terms).
- **Mentorship:** Code Coach (The Coder School) – Teaching Python, Java, C++ & Directing Summer Camps.

EXPERIENCE

AI Systems Engineer Oct 2025 – Present
Talent Strategy Experts Remote

- Sole engineer driving the migration of a low-code prototype to the production **Neuerraa** SaaS Platform; architected a secure multi-tenant core (React, Fly.io) with hierarchical RBAC and org-level data isolation.
- Engineered the full-stack MVP and public landing page, integrating intelligent AI recruiting agents while establishing CI/CD workflows and security protocols for scalable infrastructure.

Founding Engineer Sep 2025 – Present
ArkBoosted LLC Remote

- Technical Lead spearheading the architecture of AI-based web solutions, deploying scalable end-to-end systems with a security-first development lifecycle for client businesses.
- **Key Client Project (MOHID):** Architected a pgvector-backed RAG agent for a 700+ organization network, unifying knowledge retrieval to automate intake parsing, email responses, and support escalations.
- **Internal Tooling:** Engineered an AI-driven auditor (FastAPI, React) that automates SEO/security analysis, cutting internal workflow time by 90%.

LLM Evaluation Specialist (Contract) Sep 2025 – Present
Alignerr Remote

- Executed 4+ large-scale LLM evaluation initiatives spanning reasoning, coding, and function-calling domains.
- Led rubric-based and human-preference assessments (RLHF) on thousands of samples to enhance model accuracy.

ML Red Team Fellow Sep 2025 – Present
Handshake MOVE Fellowship Remote

- Red-teamed LLMs by performing penetration testing with 200+ adversarial prompts, exposing vulnerabilities, logging 50+ exploits, and delivering mitigation strategies.

Deep Learning/AI Intern Jun 2024 – Aug 2024
UIC Department of Electrical & Computer Engineering Chicago, IL

- Led a team of 5 interns in data collection and fine-tuning of PyTorch models for wildfire image classification.
- Designed "Vision Mamba", a custom CNN improving accuracy by 30% over baselines; selected for hls4ML (high level synthesis) training to convert PyTorch models for FPGA hardware acceleration.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, PHP, Bash, ARM Assembly, MATLAB, Verilog

Technologies: PyTorch, TensorFlow, OpenCV, React, Node.js, FastAPI, Docker, Kubernetes, AWS, Linux, Nginx

Hardware/Tools: MySQL, REST APIs, FPGA (Vivado), Tiva C, Raspberry Pi, LTspice, Wireshark, Valgrind, Git

PROJECTS

Full-Stack MLS Property Platform (Pro-bono) | PHP, MySQL, Cron, Linux [Live Subdomain]

- Selected as Team Lead for 6 interns at **IDXExchange**; engineered the complete backend data pipeline from scratch for my live production implementation (akbar.californsale.org) integrating CRMLS data via the Trestle API.
- Wrote PHP scripts and cron jobs to manage hourly API authentication and populate a custom MySQL schema (1,000+ live listings) while managing production deployments on Linux VPS.

MATRIX-AI Voice Assistant | Python, ElevenLabs API, WebSocket, Neural AI [Talk to Matrix]

- Developed a conversational AI assistant with real-time speech processing using Python and the ElevenLabs API; deployed as an interactive voice agent on my portfolio handling live, contextual conversations.

Real-Time Embedded Sensor System | C, ARM Assembly, Tiva C (Cortex-M4F) [GitHub]

- Engineered a bare-metal, interrupt-driven safety system featuring **dual-mode operation** (Ultrasonic Ranging vs. Thermal Monitoring); utilized Timer Capture ISRs for distance detection & ADC sequencers for heat sensing.
- Developed the **Ultra Heat Sensor** with mixed C/ARM Assembly drivers to manage hardware constraints, implementing direct register manipulation for PWM audio gen, I2C display rendering, and WFI power optimization.